Minimal Surfaces

Fall 2026 — Scuola Normale Superiore, Pisa

Giada Franz

27 October 2026

These are minimalist transcriptions of the lecture notes of a 40 hours course I am currently teaching in Scuola Normale Superiore in Pisa. The transcription of the handwritten notes into LaTeX is currently partially automated using a LLM and revised by Andrea Rocca. Work in progress.

Contents

1 Introduction **1**
- 1.1 The Plateau problem and its history . . . 2
- 1.2 Why study minimal surfaces? . . . 2
- 1.3 Course outline . . . 3

2 Preliminaries in Riemannian geometry **3**
- 2.1 Smooth manifolds, tangent spaces, and vector fields . . . 3
- 2.2 Tensors and differential forms . . . 4
- 2.3 Riemannian metrics, distance, and volume . . . 5
- 2.4 The Levi–Civita connection . . . 5
- 2.5 Curvature . . . 6
- 2.6 Operators on manifolds . . . 7
- 2.7 Submanifolds . . . 7
- 2.8 Geodesics . . . 9

3 The minimal surface equation **11**
- 3.1 Examples in $\mathbb{R}^3$ . . . 12
- 3.2 The minimal surface equation for graphs . . . 13
- 3.3 First consequences of minimality . . . 13
- 3.4 Monotonicity formula . . . 14

1 Introduction

This course aims to give an introduction to minimal surface theory, up to covering topics of current research. The content is inspired by many different references, main ones are the book by Colding–Minicozzi [CM11] and the notes by White [Whi16] and Chodosh [Cho25].

A classical motivating picture for studying minimal surfaces is a wire immersed in soapy water: after withdrawing it (and neglecting gravity), the soap film is expected to minimize its area among surfaces spanning the same boundary. Denote the wire by a smooth *Jordan curve* Γ (a simple closed curve) and a spanning surface by Σ, with $\partial\Sigma = \Gamma$.

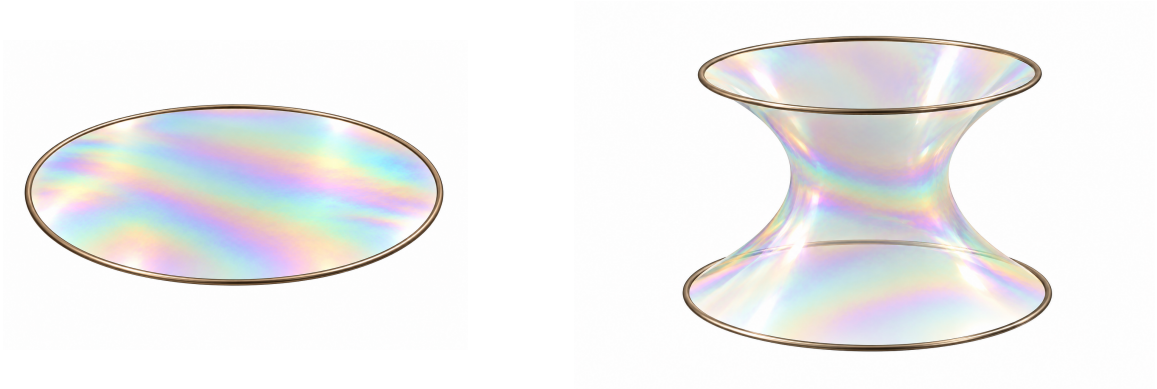


Figure 1: The Plateau problem: seek a least-area spanning surface Σ for a prescribed boundary curve Γ. A disk and a catenoid are familiar examples of minimal surfaces.

1.1 The Plateau problem and its history

Finding a surface with least area has a long history. In 1762, Lagrange [Lag67] wrote down the equation for graphs of least area (the original memoir appeared in the volume dated 1760–1761). The problem of minimizing area with a fixed boundary curve Γ is named after Joseph Plateau, who investigated soap films (his findings were published in 1873 [Pla73], after decades of research).

The classical Plateau problem was solved in the 1930s, independently by Jesse Douglas and Tibor Radó [Dou31, Rad30]. Douglas received one of the first Fields Medals in 1936 for this work. An area-minimizing surface need not initially be given as an embedding: the construction may yield immersed surfaces and require an analysis of possible singularities and branch points.

For boundary curves in $\mathbb{R}^3$, work of Osserman and Gulliver in the 1970s ruled out interior branch points of area-minimizing disks [Oss70, Gul73]. When $\Gamma \subset \partial K \subset \mathbb{R}^3$ for a compact convex set K, Meeks–Yau [MY82] in 1982 established embeddedness results for least-area solutions.

Open problem. *It is an open problem whether branch points can occur at the (smooth) boundary.*

Remark 1.1 (Branched minimal surfaces). The graph of the map $F\colon \mathbb{C} \to \mathbb{C}^2$ given by

$$F(z) = (z^2, z^3) \in \mathbb{C}^2$$

is a branched minimal surface.

Morrey [Mor48] in 1948 generalized the problem to surfaces in Riemannian manifolds (M, g). There is also a higher-dimensional theory of area minimization. In the setting of geometric measure theory, the Federer–Fleming theory (1960) [FF60] provides least-mass representatives for appropriate integral homology classes, schematically

$$\sigma \in H_k(M; \mathbb{Z}) \quad \Longrightarrow \quad \text{an area-minimizing integral } k\text{-cycle representing } \sigma.$$

For codimension-one minimizers, write $\Sigma^n \subset M^{n+1}$. When $n \leq 6$ (ambient dimension at most 7), regularity theory yields smooth area-minimizing hypersurfaces in the interior, under the usual hypotheses; see also the foundational regularity work of Simons [Sim68].

The subject subsequently developed in many directions: finding area-minimizing surfaces in new settings, constructing surfaces that are only critical points of the area functional (*minimal surfaces*), and understanding their geometry and topology.

1.2 Why study minimal surfaces?

Minimal surfaces are beautiful geometric objects in their own right. They also arise naturally in many problems as critical points or minimizers of geometric energies. Some examples are:

- **Scalar curvature and rigidity:** which Riemannian metrics on a torus T^n can have nonnegative scalar curvature? [SY79a]
- **General relativity:** the positive mass theorem [SY79b].
- **Knot theory:** a closed curve in $\mathbb{R}^3$ with total curvature less than 4π is unknotted [Mil50]; counting minimal disks in hyperbolic 4-space $\mathbb{H}^4$ with prescribed knot as ideal boundary is related to knot invariants.
- **Image processing:** image denoising and optimal transport.
- **3-manifolds:** finding useful splittings, especially Heegaard splittings.
- **The Smale conjecture and diffeomorphism groups:** studying $\mathrm{Diff}(M)$ for a 3-manifold M.
- **Spectral theory.**

Some applications will be studied in the lectures; others are proposed as possible exam topics.

1.3 Course outline

1. Preliminaries in Riemannian geometry.
2. Foundations of minimal surface theory, including minimal surfaces in Euclidean space.
3. Minimal surfaces in positive curvature.
4. Minimal surfaces in negative curvature.
5. Existence of minimal surfaces.

2 Preliminaries in Riemannian geometry

2.1 Smooth manifolds, tangent spaces, and vector fields

Let M^n be a smooth manifold and $p \in M$. Its tangent space T_pM may be defined as the vector space of velocities $\gamma'(0)$ of smooth curves γ passing through p. Equivalently, it consists of derivations at p, i.e., linear maps

$$v\colon C^\infty(M) \longrightarrow \mathbb{R}, \qquad v(fg) = f(p)v(g) + g(p)v(f).$$

Under this identification, the tangent vector $v = \gamma'(0)$ acts by

$$\gamma'(0)(f) = \frac{\mathrm{d}}{\mathrm{d}t}\bigg|_{t=0} f(\gamma(t)).$$

The *tangent bundle* is the disjoint union

$$TM = \bigsqcup_{p\in M} T_pM.$$

A chart $(x^1, \dots, x^n)\colon U \subset M \to \mathbb{R}^n$ induces a basis $\left\{\partial_i|_p = \frac{\partial}{\partial x^i}\Big|_p\right\}_{i=1}^n$ of T_pM.

For a smooth map $F : M \to N$ between smooth manifolds, the *differential* at p is

$$\mathrm{d}F_p\colon T_pM \longrightarrow T_{F(p)}N, \qquad (\mathrm{d}F_p(v))(f) = v(f \circ F).$$

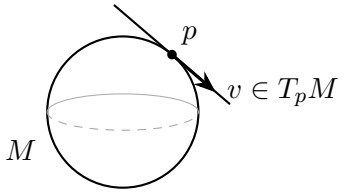


Figure 2: Schematic view of a tangent vector at a point of a manifold.

It satisfies the chain rule

$$\mathrm{d}(G \circ F)_p = \mathrm{d}G_{F(p)} \circ \mathrm{d}F_p.$$

The *cotangent space* is the dual $T_p^*M = (T_pM)^*$, and the *cotangent bundle* is T^*M. For $f \in C^\infty(M)$,

$$\mathrm{d}f_p \in T_p^*M, \qquad \mathrm{d}f_p(v) = v(f).$$

In local coordinates, the covectors $\mathrm{d}x^i|_p$ form the dual basis to the $\partial_i|_p$ and

$$\mathrm{d}f = \frac{\partial f}{\partial x^i}\mathrm{d}x^i.$$

Here and below, repeated upper and lower indices may be summed over, in Einstein notation.

A *smooth vector field* $X \in \mathfrak{X}(M) = \Gamma(TM)$ is a smooth section of the tangent bundle. Locally,

$$X = X^i\partial_i, \qquad X(f) = \mathrm{d}f(X).$$

Every smooth vector field has a local *flow* Φ_t^X, defined by

$$\frac{\partial}{\partial t}\Phi_t^X(p) = X(\Phi_t^X(p)), \qquad \Phi_0^X(p) = p.$$

If M is compact, or more generally if X has compact support, the flow is defined for all $t \in \mathbb{R}$, and each $\Phi_t^X \colon M \to M$ is a diffeomorphism.

2.2 Tensors and differential forms

An (r, s)-tensor field is a $C^\infty(M)$-multilinear map

$$T\colon \underbrace{\Gamma(T^*M) \times \cdots \times \Gamma(T^*M)}_{r \text{ factors}} \times \underbrace{\Gamma(TM) \times \cdots \times \Gamma(TM)}_{s \text{ factors}} \longrightarrow C^\infty(M).$$

We denote by $T^{(r,s)}(M)$ the space of (r, s)-tensor fields. In coordinates,

$$T = T^{i_1\cdots i_r}_{j_1\cdots j_s}\,\partial_{i_1} \otimes \cdots \otimes \partial_{i_r} \otimes \mathrm{d}x^{j_1} \otimes \cdots \otimes \mathrm{d}x^{j_s}.$$

In particular, vector fields are $(1, 0)$-tensor fields.

A *k-form* $\omega \in \Omega^k(M)$ is an alternating $(0, k)$-tensor. Thus

$$\omega(\ldots, X_i, \ldots, X_j, \ldots) = -\omega(\ldots, X_j, \ldots, X_i, \ldots).$$

The *wedge product* is the alternating tensor product: for $\alpha \in \Omega^k(M)$ and $\beta \in \Omega^\ell(M)$,

$$(\alpha \wedge \beta)(X_1, \ldots, X_{k+\ell}) = \frac{1}{k!\ell!}\sum_{\sigma \in S_{k+\ell}} \operatorname{sgn}(\sigma)\alpha(X_{\sigma(1)}, \ldots, X_{\sigma(k)})\beta(X_{\sigma(k+1)}, \ldots, X_{\sigma(k+\ell)}).$$

A manifold of dimension n is *orientable* if it admits a nowhere-vanishing smooth n-form.

2.3 Riemannian metrics, distance, and volume

A *Riemannian manifold* (M^n, g) is a smooth manifold endowed with a smooth, symmetric, positive-definite $(0,2)$-tensor g, its Riemannian metric. It defines an inner product $\langle \cdot, \cdot \rangle_p$ on each T_pM. Locally,

$$g = g_{ij}\,\mathrm{d}x^i \otimes \mathrm{d}x^j, \qquad (g^{ij}) = (g_{ij})^{-1}.$$

The *Riemannian distance* is the infimum of lengths of piecewise smooth curves joining two points:

$$d(p,q) = \inf_{\substack{\gamma\colon [a,b]\to M \\ \gamma(a)=p,\,\gamma(b)=q}} \ell(\gamma), \qquad \ell(\gamma) = \int_a^b |\gamma'(t)|_g \mathrm{d}t.$$

If M is oriented, its *Riemannian volume form* $\mathrm{d}\,\mathrm{vol}_g$ takes value 1 on each positively oriented orthonormal basis $e_1, \dots, e_n$. In oriented coordinates,

$$\mathrm{d}\,\mathrm{vol}_g = \sqrt{\det(g_{ij})}\,\mathrm{d}x^1 \wedge \cdots \wedge \mathrm{d}x^n,$$

so integration becomes

$$\int_M f\,\mathrm{d}\,\mathrm{vol}_g = \int_{x(U)} (f \circ x^{-1}) \sqrt{\det(g_{ij})}\,\mathrm{d}x^1 \cdots \mathrm{d}x^n$$

when f is supported inside a positively oriented coordinate chart U.

Theorem 2.1 (Stokes' theorem)**.** *For a smooth oriented n-manifold M (possibly with boundary) and a compactly supported $(n-1)$-form ω,*

$$\int_M \mathrm{d}\omega = \int_{\partial M} \omega,$$

using the induced orientation on ∂M.

2.4 The Levi–Civita connection

A *connection* on the tangent bundle assigns to vector fields X, Y a new vector field $\nabla_X Y$. The *Levi–Civita connection* is characterized by the following conditions:

(i) ∇ is $C^\infty(M)$-linear in X, $\mathbb{R}$-linear in Y, and satisfies the Leibniz rule

$$\nabla_X(fY) = X(f)Y + f\nabla_X Y.$$

(ii) *Torsion-free:*

$$\nabla_X Y - \nabla_Y X = [X, Y].$$

(iii) *Metric compatible:*

$$X\big(g(Y,Z)\big) = g(\nabla_X Y, Z) + g(Y, \nabla_X Z).$$

In local coordinates the *Christoffel symbols* Γ_{ij}^k are defined by

$$\nabla_{\partial_i} \partial_j = \Gamma_{ij}^k \partial_k.$$

On Euclidean space $\mathbb{R}^n$, the Levi–Civita connection is the ordinary directional derivative:

$$\nabla^{\mathbb{R}^n}_X Y = D_X Y = \left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0} Y(p + tX(p)).$$

Along a curve $\gamma\colon [a,b] \to M$, covariant differentiation of a vector field Y along γ is written

$$\frac{\nabla Y}{\mathrm{d}t} = \nabla_{\gamma'(t)} Y.$$

Following the notation in the handwritten notes, for a vector field Y *along* γ we abbreviate its first and second covariant derivatives by

$$Y' := \frac{\nabla Y}{\mathrm{d}t}, \qquad Y'' := \frac{\nabla}{\mathrm{d}t} Y'.$$

Thus a prime on a vector field along γ means *covariant* differentiation; in $\mathbb{R}^n$ this agrees with the usual derivative. The connection also extends, via the Leibniz rule, to tensor fields.

2.5 Curvature

The *Riemann curvature tensor* R of (M^n, g) is defined by

$$R(X,Y)Z := \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X,Y]} Z.$$

We also use the associated $(0,4)$-tensor, with the following convention for the last two arguments:

$$R(X,Y,Z,W) := g(R(X,Y)W, Z).$$

The *sectional curvature* of the plane spanned by linearly independent vectors $X, Y \in T_pM$ is

$$\operatorname{Sec}(X,Y) = \frac{R(X,Y,X,Y)}{|X|^2|Y|^2 - g(X,Y)^2}.$$

The *Ricci curvature* is the trace

$$\operatorname{Ric}(X,Y) = \sum_{i=1}^{n} R(X, e_i, Y, e_i),$$

where $(e_i)_{i=1}^n$ is an orthonormal basis of T_pM. The *scalar curvature* is

$$\operatorname{Scal} = \operatorname{tr}_g \operatorname{Ric} = \sum_{i \neq j} \operatorname{Sec}(e_i, e_j).$$

The curvature tensor satisfies the following symmetries and identities:

$$\begin{aligned} R(X,Y,Z,W) &= R(Z,W,X,Y) = -R(X,Y,W,Z), \\ R(X,Y)Z + R(Y,Z)X + R(Z,X)Y &= 0 \qquad \text{(first Bianchi identity)}, \\ (\nabla_X R)(Y,Z) + (\nabla_Y R)(Z,X) + (\nabla_Z R)(X,Y) &= 0 \qquad \text{(second Bianchi identity)}. \end{aligned}$$

In the last identity, R is viewed as a $(1,3)$-tensor and the equality holds after applying both sides to a fourth vector.

Example 2.2 (Space forms). The model spaces $\mathbb{R}^n$, S^n (round unit sphere), and $\mathbb{H}^n$ have constant sectional curvature $\kappa = 0, 1, -1$, respectively. More generally, on a manifold of constant sectional curvature κ,

$$\operatorname{Ric} = (n-1)\kappa\, g, \qquad \operatorname{Scal} = n(n-1)\kappa.$$

These models are, up to isometry, the complete simply connected space forms in their respective curvature classes.

2.6 Operators on manifolds

The metric induces the *musical isomorphisms*

$$\begin{aligned} \flat\colon TM &\longrightarrow T^*M, \qquad X^\flat(Y) = g(X,Y), \\ \sharp\colon T^*M &\longrightarrow TM, \qquad g(\omega^\sharp, Y) = \omega(Y). \end{aligned}$$

In coordinates,

$$(X^\flat)_j = g_{ij}X^i, \qquad (\omega^\sharp)^i = g^{ij}\omega_j.$$

For $f \in C^\infty(M)$ and $X \in \mathfrak{X}(M)$, we define

$$\begin{aligned} \nabla f &:= (\mathrm{d}f)^\sharp && \text{(gradient)}, \\ \operatorname{div} X &:= \operatorname{tr}(\nabla X) = \sum_{i=1}^n g(\nabla_{e_i} X, e_i) && \text{(divergence)}, \\ \nabla^2 f &:= \nabla(\mathrm{d}f), \quad \nabla^2 f(X,Y) = g(\nabla_X \nabla f, Y) && \text{(Hessian)}, \\ \Delta f &:= \operatorname{div}(\nabla f) = \operatorname{tr}_g(\nabla^2 f) && \text{(Laplacian)}. \end{aligned}$$

Here (e_i) is a local orthonormal frame. In particular, the convention for Δ is $\Delta = \operatorname{div}\nabla$.

Theorem 2.3 (Divergence theorem)**.** *Let (M^n, g) be oriented, with smooth boundary, and let X be a compactly supported smooth vector field (smooth up to the boundary). If ν is the outward-pointing unit normal to ∂M, then*

$$\int_M \operatorname{div} X \,\mathrm{d}\operatorname{vol}_g = \int_{\partial M} g(X,\nu)\,\mathrm{d}\operatorname{vol}_{\partial M}.$$

Proof. By Cartan's formula for the Lie derivative and the fact that a top-degree form is closed,

$$\mathrm{d}(\iota_X \mathrm{d}\operatorname{vol}_g) = \mathcal{L}_X \mathrm{d}\operatorname{vol}_g = (\operatorname{div} X)\,\mathrm{d}\operatorname{vol}_g.$$

Moreover, on ∂M one has

$$(\iota_X \mathrm{d}\operatorname{vol}_g)|_{\partial M} = g(X,\nu)\,\mathrm{d}\operatorname{vol}_{\partial M}.$$

The assertion follows from Stokes' theorem. □

Remark 2.4. An alternative proof in local coordinates, using the divergence formula and Gauss's theorem in $\mathbb{R}^n$, is left as an exercise.

2.7 Submanifolds

We use $\Sigma^k \subset M^n$ for a k-dimensional submanifold of an n-dimensional ambient manifold. For hypersurfaces we instead write $\Sigma^n \subset M^{n+1}$.

For a k-dimensional submanifold S^k, we use $\operatorname{vol}(S)$ and $|S|$ interchangeably for its k-dimensional Riemannian volume (measure). Thus $|\Sigma|$ denotes k-dimensional volume when $\Sigma^k \subset M^n$, whereas $|\partial\Sigma|$ denotes the $(k-1)$-dimensional measure of its boundary.

For a smooth map $F\colon \Sigma^k \to M^n$:

- F is an *immersion* if $\mathrm{d}F_p$ is injective for every $p \in \Sigma$;
- F is an *embedding* if it is an immersion and a homeomorphism onto its image (with the subspace topology);
- F is an *isometry* if $F^* g_M = g_\Sigma$, where g_Σ, g_M are metrics on Σ, M.

From now on, unless otherwise stated, we assume that $F\colon \Sigma^k \to M^n$ is an isometric immersion. As usual, we identify Σ with its image locally and regard tangent vector fields as fields along the immersion. The intrinsic Levi–Civita connection is the tangential projection of the ambient one:

$$\nabla^{\Sigma}_X Y = (\nabla^{M}_X Y)^{\top}.$$

In this subsection, for brevity, write $\nabla := \nabla^{\Sigma}$ and $D := \nabla^{M}$.

Definition 2.5 (Second fundamental form)**.** The *second fundamental form* is the normal-bundle-valued symmetric bilinear form

$$A(X,Y) := (D_X Y)^{\perp} \in T\Sigma^{\perp}.$$

For tangent fields X, Y, it satisfies

$$A(X,Y) = A(Y,X), \qquad D_X Y = \nabla_X Y + A(X,Y).$$

The (unnormalized) *mean curvature vector* is

$$\vec{H} := \operatorname{tr}_{T\Sigma} A = \sum_{i=1}^{k} A(e_i, e_i).$$

Proposition 2.6 (Gauss equation)**.** *For tangent vectors $X, Y, Z, W \in T_p\Sigma$,*

$$R^{\Sigma}(X,Y,Z,W) = R^{M}(X,Y,Z,W) + g(A(X,Z),A(Y,W)) - g(A(X,W),A(Y,Z)).$$

Proposition 2.7 (Codazzi equation)**.** *For tangent fields X, Y, Z,*

$$(\nabla_X A)(Y,Z) - (\nabla_Y A)(X,Z) = \big(R^{M}(X,Y)Z\big)^{\perp},$$

where ∇A uses the induced tangent and normal connections.

Hypersurfaces

For a hypersurface $\Sigma^n \subset M^{n+1}$, we say Σ is *two-sided* if there exists a global unit normal field $N\colon \Sigma \to T\Sigma^{\perp}$. Once N is chosen, define

$$\mathrm{II}(X,Y) := -g(A(X,Y),N), \qquad H := -g(\vec{H},N).$$

Thus $H = \operatorname{tr}_{T\Sigma} \mathrm{II}$.

Example 2.8 (Round sphere). For $S^n \subset \mathbb{R}^{n+1}$ with the outward unit normal $N(x) = x$,

$$A(X,Y) = -g(X,Y)N, \qquad \mathrm{II}(X,Y) = g(X,Y),$$

and hence $\vec{H} = -nN$ and $H = n$.

Definition 2.9 (Shape operator)**.** For a chosen unit normal N, the *shape operator* is

$$S_N\colon T_p\Sigma \longrightarrow T_p\Sigma, \qquad S_N(X) := -(D_X N)^{\top}.$$

It satisfies

$$g(S_N(X),Y) = g(A(X,Y),N) = -\mathrm{II}(X,Y).$$

The *principal curvatures* (with this convention) are the eigenvalues of S_N.

Remark 2.10. With the signs above, the eigenvalues of S_N are the negatives of those of the quadratic form II with respect to g. In particular, $S_N = -\operatorname{id}$ for the outward-oriented unit sphere.

2.8 Geodesics

Let (M^n, g) be a complete Riemannian manifold (for example, $S^3 \subset \mathbb{R}^4$ with the round metric). Given $p, q \in M$, what is the shortest path between them?

For a piecewise smooth curve $\gamma\colon I = [a,b] \to M$, define its *length* by

$$\ell(\gamma) = \int_a^b |\gamma'(t)| \mathrm{d}t, \qquad |\gamma'(t)| = \sqrt{g(\gamma'(t), \gamma'(t))}.$$

Given two points p, q, we seek a curve γ between them minimizing $\ell(\gamma)$.

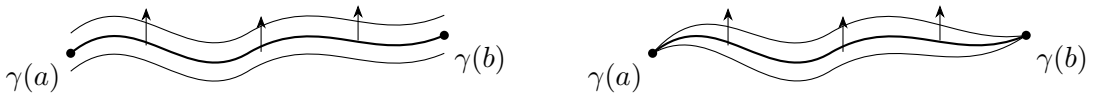
It is often convenient to consider the *energy*

$$E(\gamma) = \frac{1}{2} \int_a^b |\gamma'(t)|^2 \mathrm{d}t.$$

By Cauchy–Schwarz, $\ell(\gamma)^2 \leq 2(b-a)E(\gamma)$, with equality if and only if γ has constant speed. A diffeomorphism $\sigma\colon I \to I$ changes the parametrization of γ without changing its image, so a regular curve can be reparametrized at constant speed. Thus energy minimizers correspond to length minimizers parametrized with constant speed.

Definition 2.11. A smooth map $F\colon I \times (-\varepsilon, \varepsilon) \to M$ is a *variation* of γ if $F(t,0) = \gamma(t)$. It is *proper* (fixed endpoints) if $F(a,s) = \gamma(a)$ and $F(b,s) = \gamma(b)$ for every s. Its associated *variational field* is

$$X(t) = \left.\frac{\partial F(t,s)}{\partial s}\right|_{s=0}.$$



Variation with moving endpoints — Proper variation (fixed endpoints)

Figure 3: Two variations of a curve. On the left the endpoints move; on the right they remain fixed (a proper variation).

Proposition 2.12 (First variation of energy)**.** *For a variation F of γ with variational field X,*

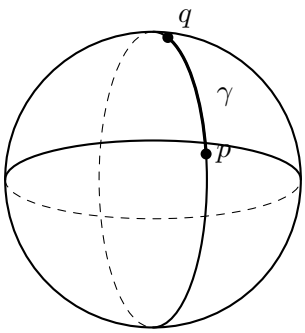
$$\left.\frac{\mathrm{d}}{\mathrm{d}s} E(F(\cdot, s))\right|_{s=0} = -\int_a^b \left\langle X, \frac{\nabla}{\mathrm{d}t}\gamma' \right\rangle \mathrm{d}t + \langle X(b), \gamma'(b)\rangle - \langle X(a), \gamma'(a)\rangle.$$

Definition 2.13. A smooth curve $\gamma\colon I \to M$ is a *geodesic* if $\frac{\nabla}{\mathrm{d}t}\gamma' = 0$ on I. Equivalently, it is a critical point of the energy with respect to proper variations. A *(minimizing) geodesic segment* is a geodesic which minimizes the length between its endpoints.

Example 2.14 (Geodesics on S^2). Great circles are geodesics on the round sphere. A long arc of a great circle need not be length minimizing between its endpoints.

Remark 2.15. Every minimizing segment is a geodesic, but the converse need not hold globally. Every geodesic is locally a minimizing segment.

Proposition 2.16 (Local existence and uniqueness)**.** *For every $p \in M$, there is a neighborhood U_p of $0 \in T_pM$ such that for any $v \in U_p$, there is a unique geodesic starting at p with initial velocity v, defined at least until time 1. The exponential map is* $\exp_p(v) = \gamma_v(1)$.



γ is a geodesic segment: an arc of a great circle on the round sphere.

Figure 4: The round sphere S^2 with a geodesic γ joining two points p and q. Great circles are exactly the geodesics of the round metric.

Proposition 2.17 (Second variation of energy)**.** *Suppose γ is a geodesic and F is a variation of γ with variational field X. With the curvature convention used in these notes,*

$$\begin{aligned}\frac{\mathrm{d}^2}{\mathrm{d}s^2}E(F(\cdot,s))\Big|_{s=0} &= \int_a^b (|X'|^2 - \langle R(X,\gamma')\gamma', X\rangle)\mathrm{d}t + \left[\langle \nabla_s \partial_s F|_{s=0}, \gamma'\rangle\right]_a^b \\ &= -\int_a^b \langle X, X'' + R(X,\gamma')\gamma'\rangle \,\mathrm{d}t + \langle X, X'\rangle\big|_a^b + \left[\langle \nabla_s \partial_s F|_{s=0}, \gamma'\rangle\right]_a^b .\end{aligned}$$

For a proper variation both boundary terms vanish.

Definition 2.18. The *Jacobi operator* along a geodesic is

$$J(X) = X'' + R(X,\gamma')\gamma'.$$

A field in the kernel of J is called a *Jacobi field*.

Theorem 2.19 (Myers [Mye41])**.** *Let (M^n, g) be complete and suppose*

$$\mathrm{Ric} \geq (n-1)\kappa\, g \qquad \text{for some } \kappa > 0.$$

Then (i) M is compact, with $\operatorname{diam}(M) \leq \pi/\sqrt{\kappa}$; and (ii) $\pi_1(M)$ is finite.

Proof. For (i), let $\gamma\colon [0,\ell] \to M$ be a unit-speed geodesic. It suffices to show that if $\ell > \pi/\sqrt{\kappa}$, then $\gamma|_{[0,\ell]}$ is not minimizing.

Choose an orthonormal basis $E_\alpha(0)$, $\alpha = 1, \dots, n-1$, of $\gamma'(0)^\perp \subset T_pM$, and parallel-transport[1] it along γ; thus $E'_\alpha = 0$. Set $c = (\pi/\ell)^2 < \kappa$ and define

$$\mathrm{sn}_c(t) = \frac{1}{\sqrt{c}}\sin(\sqrt{c}\,t), \qquad V_\alpha(t) = \mathrm{sn}_c(t)E_\alpha(t), \qquad \mathrm{sn}_c'' + c\,\mathrm{sn}_c = 0.$$

The proper variations $F_\alpha(t,s) = \exp_{\gamma(t)}(sV_\alpha(t))$ have second variations whose sum equals

$$\begin{aligned}\sum_{\alpha=1}^{n-1}\frac{\mathrm{d}^2}{\mathrm{d}s^2}E(F_\alpha(\cdot,s))\Big|_{s=0} &= \sum_{\alpha=1}^{n-1}\int_0^\ell \Big(|V_\alpha'|^2 - \langle R(V_\alpha,\gamma')\gamma', V_\alpha\rangle\Big)\,\mathrm{d}t \\ &= \int_0^\ell \Big((n-1)(\mathrm{sn}_c')^2 - \mathrm{sn}_c^2\,\mathrm{Ric}(\gamma',\gamma')\Big)\,\mathrm{d}t \\ &\leq (n-1)\int_0^\ell \big((\mathrm{sn}_c')^2 - \kappa\,\mathrm{sn}_c^2\big)\mathrm{d}t \\ &= (n-1)\int_0^\ell (c-\kappa)\,\mathrm{sn}_c^2\,\mathrm{d}t < 0.\end{aligned}$$

[1]Namely, $E'_\alpha(t) = \frac{\nabla}{dt}E_\alpha = 0$. Writing it down in coordinates will actually give a ODE.

Here integration by parts and $\mathrm{sn}_c(0) = \mathrm{sn}_c(\ell) = 0$ give $\int (\mathrm{sn}_c')^2 = c\int \mathrm{sn}_c^2$. Thus at least one proper variation has negative second variation, contradicting minimizing behavior. Completeness and Hopf–Rinow now imply compactness and the diameter bound.

For (ii), let $\pi : \widetilde{M} \to M$ be the universal cover equipped with the pullback metric $\widetilde{g} = \pi^* g$. The cover is complete and satisfies $\widetilde{\mathrm{Ric}} \geq (n-1)\kappa \widetilde{g}$. By (i), $\widetilde{M}$ is compact. A compact universal cover has only finitely many sheets over M, so $\pi_1(M)$ is finite. □

Remark 2.20. The theorem uses geodesics to infer global geometric and topological information about a manifold.

3 The minimal surface equation

Let (M^n, g) be a complete Riemannian manifold and $\Sigma^k \subset M^n$ an embedded submanifold. (Much of the discussion also applies to immersed submanifolds.)

Definition 3.1. For a smooth vector field X along Σ, define its divergence along Σ by

$$\mathrm{div}_\Sigma X = \sum_{i=1}^{k} g(\nabla_{e_i} X, e_i) = \mathrm{tr}_{T\Sigma}(\nabla X),$$

where $\{e_1, \dots, e_k\}$ is an orthonormal basis of $T_p\Sigma$. One can compute this using an extension of X to a neighborhood in M; the result is independent of the extension.

Definition 3.2. A *compactly supported variation* of Σ is a smooth map $F\colon \Sigma \times (-\varepsilon, \varepsilon) \to M$ such that $F(x,0) = x$ and $F(x,t) = x$ outside a compact subset of Σ. Its variational field is

$$X(x) = \left.\frac{\partial F(x,t)}{\partial t}\right|_{t=0}.$$

Proposition 3.3 (First variation of volume)**.** *The first variation of the k-dimensional volume of Σ is*

$$\delta\Sigma(X) := \left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0} \mathrm{vol}(F_t(\Sigma)) = \int_\Sigma \mathrm{div}_\Sigma X \,\mathrm{d}\,\mathrm{vol}_\Sigma = -\int_\Sigma \langle \vec{H}, X\rangle \mathrm{d}\,\mathrm{vol}_\Sigma + \int_{\partial\Sigma} \langle X, \eta\rangle \mathrm{d}\,\mathrm{vol}_{\partial\Sigma},$$

where $\vec{H}$ is the (unnormalized) mean curvature vector and η is the outward pointing unit conormal along $\partial\Sigma$.

Proof. Choose local coordinates $x^1, \dots, x^k$ on Σ and write

$$g_{ij}(t) = g(\partial_i F, \partial_j F), \qquad \mathrm{d}\,\mathrm{vol}_{F_t(\Sigma)} = \sqrt{\det g_{ij}(t)}\,\mathrm{d}x^1 \cdots \mathrm{d}x^k.$$

The ratio of induced volume densities is

$$v(t) \coloneqq \frac{\mathrm{d}\,\mathrm{vol}_{F_t(\Sigma)}}{\mathrm{d}\,\mathrm{vol}_\Sigma} = \frac{\sqrt{\det g_{ij}(t)}}{\sqrt{\det g_{ij}(0)}}.$$

At a point $p \in \Sigma$ choose coordinates orthonormal at $t = 0$, so $g_{ij}(0) = \delta_{ij}$. Then

$$\begin{aligned}\left.\frac{\mathrm{d}}{\mathrm{d}t} v(t)\right|_0 &= \frac{1}{2}\left.\frac{\mathrm{d}}{\mathrm{d}t} \det g_{ij}(t)\right|_0 = \frac{1}{2}\sum_{i=1}^{k} \left.\frac{\mathrm{d}}{\mathrm{d}t} g_{ii}(t)\right|_0 \\ &= \sum_{i=1}^{k} \langle \nabla_{\partial_t F} \partial_i F, \partial_i F\rangle = \sum_{i=1}^{k} \langle \nabla_{\partial_i F} \partial_t F, \partial_i F\rangle = \mathrm{div}_\Sigma X.\end{aligned}$$

Split $X = X^\perp + X^\top$. For an orthonormal tangent frame,

$$\operatorname{div}_\Sigma X^\perp = \sum_{i=1}^{k} \langle \nabla_{e_i} X^\perp, e_i \rangle = -\sum_{i=1}^{k} \langle X^\perp, \nabla_{e_i} e_i \rangle = -\langle X, \vec{H} \rangle,$$
$$\operatorname{div}_\Sigma X^\top = \operatorname{div}_\Sigma^{\text{intrinsic}} X^\top.$$

The intrinsic divergence theorem now yields the assertion. □

Definition 3.4. A submanifold $\Sigma \subset M$ is *minimal* if its mean curvature vector $\vec{H}$ vanishes identically. Equivalently, Σ is a critical point of the area functional for every compactly supported variation fixing $\partial\Sigma$.

Remark 3.5. Area minimizers are minimal, but minimal submanifolds need not globally minimize area.

3.1 Examples in $\mathbb{R}^3$

- **Plane:** $\Sigma = \{z = 0\} \subset \mathbb{R}^3$ has $\vec{H} = 0$.
- **Disk:** $\Sigma = \{(x, y, 0) : x^2 + y^2 \leq 1\}$ is minimal with boundary. For compactly supported variations and constant vertical displacement $X = (0, 0, 1)$ the first variation is zero; an outward radial variation $X = (x, y, 0)$ changes its area, owing to the boundary term:

$$\delta\Sigma(X) = 2\pi = \frac{\mathrm{d}}{\mathrm{d}t}\Big|_{t=0} \pi(1+t)^2.$$

- **Catenoid:** A surface of revolution, for instance $x^2 + y^2 = \cosh^2 z$. Two sufficiently separated coaxial circles cannot be joined by an area-minimizing catenoid; compare with two disks. Discovered by Euler (1744) [Eul44]; studied by Meusnier (1776, with a memoir published in 1785) [Meu85]. Recall $\cosh z = (e^z + e^{-z})/2$.

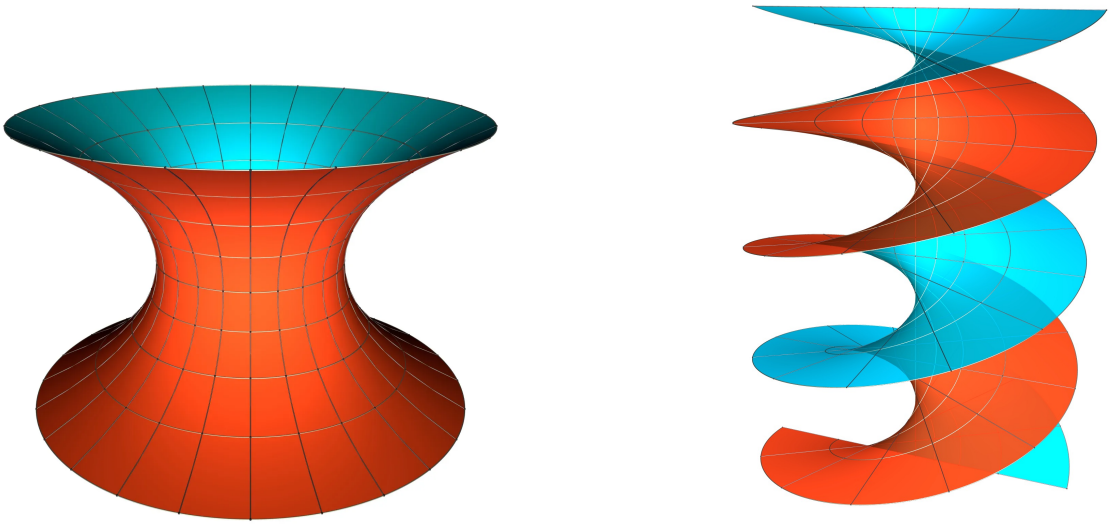


Figure 5: Catenoid (left) and helicoid (right) from the minimal surfaces gallery of Matthias Weber [Web].

- **Helicoid:** One parametrization is

$$F(t, s) = (t\cos s, t\sin s, s), \qquad s, t \in \mathbb{R}.$$

Away from the axis its equation can be expressed locally as $z = \arctan(y/x)$, modulo branches. The helicoid is a complete embedded minimal surface, simply connected, and ruled. Meusnier (1776); Catalan (1842) showed it is the only nonplanar ruled minimal surface [Cat42]. For illustrations of the helicoid, see the gallery by Matthias Weber [Web].

3.2 The minimal surface equation for graphs

For a graph $\Sigma^n = \{(x, u(x)) : x \in \Omega \subset \mathbb{R}^n\} \subset \mathbb{R}^{n+1}$, the minimal surface equation is

$$\operatorname{div}\left(\frac{\nabla u}{\sqrt{1+|\nabla u|^2}}\right) = 0.$$

In coordinates this can be written as

$$\Delta u - \frac{\sum_{i,j=1}^n u_i u_j u_{ij}}{1+|\nabla u|^2} = 0, \qquad \text{or} \qquad \sum_{i,j=1}^n \left(\delta_{ij} - \frac{u_i u_j}{1+|\nabla u|^2}\right) u_{ij} = 0.$$

This is a second-order quasilinear elliptic PDE.

3.3 First consequences of minimality

Proposition 3.6. *A submanifold $\Sigma^k \subset \mathbb{R}^n$ is minimal if and only if the restrictions of the Euclidean coordinate functions $x^1, \dots, x^n$ to Σ are harmonic.*

Proof. Let $E_j = \nabla x^j$ in $\mathbb{R}^n$ and $\varphi \in C_c^\infty(\Sigma \setminus \partial\Sigma)$. Set $X = \varphi E_j$. Since the Euclidean connection makes E_j parallel,

$$\operatorname{div}_\Sigma X = \sum_{i=1}^k \langle \nabla_{e_i}(\varphi E_j), e_i\rangle = \langle \nabla_\Sigma \varphi, E_j\rangle.$$

Therefore

$$\begin{aligned}\int_\Sigma \varphi \Delta_\Sigma x^j \,\mathrm{d}\operatorname{vol}_\Sigma &= -\int_\Sigma \langle \nabla_\Sigma \varphi, \nabla_\Sigma x^j\rangle \,\mathrm{d}\operatorname{vol}_\Sigma \\ &= -\int_\Sigma \operatorname{div}_\Sigma(\varphi E_j)\,\mathrm{d}\operatorname{vol}_\Sigma = \int_\Sigma \varphi\langle \vec{H}, E_j\rangle \,\mathrm{d}\operatorname{vol}_\Sigma .\end{aligned}$$

Thus $\Delta_\Sigma \vec{x} = \vec{H}$ and the claim follows. □

Proposition 3.7. *Let $\Sigma^k \subset \mathbb{R}^n$ be a compact minimal submanifold with boundary. Then*

$$k \operatorname{area}(\Sigma) = \int_{\partial\Sigma} \langle x, \eta\rangle \,\mathrm{d}\operatorname{vol}_{\partial\Sigma}.$$

If $\Sigma \subset B_R(0)$ and $\partial\Sigma \subset \partial B_R(0)$, then

$$\operatorname{area}(\Sigma) \le \frac{R}{k}\operatorname{area}(\partial\Sigma).$$

Proof. Take $X(x) = x$. In Euclidean space $\nabla_v X = v$, so $\operatorname{div}_\Sigma X = k$. Apply the first variation/divergence formula and $\vec{H} = 0$. □

Remark 3.8. For a minimal $\Sigma^k \subset \mathbb{R}^n$, $\Delta_\Sigma |x|^2 = 2k$.

Proposition 3.9 (Convex hull property)**.** *If $\Sigma^k \subset \mathbb{R}^n$ is a compact minimal submanifold with boundary, then*

$$\Sigma \subset \operatorname{Conv}(\partial\Sigma).$$

Proof. For every fixed unit vector v, the function $u(x) = \langle v, x\rangle$ is harmonic on Σ and attains its maximum on $\partial\Sigma$ by the maximum principle. Hence any half-space containing $\partial\Sigma$ contains Σ. Intersecting these half-spaces proves the claim. □

Corollary 3.10. *There are no compact minimal submanifolds of $\mathbb{R}^n$ without boundary (of positive dimension).*

Corollary 3.11. *Let $\Sigma^2 \subset \mathbb{R}^n$ be a compact minimal surface with boundary, homeomorphic to a disk. Let $K \subset \mathbb{R}^n$ be compact and convex with $K \cap \partial\Sigma = \emptyset$. Then $K \cap \Sigma$ is simply connected.*

Sketch. By contradiction, suppose that a closed curve $\gamma \subset K \cap \Sigma$ does not bound a disk in $K \cap \Sigma$. Since Σ is disk-like, γ bounds a disk $\Gamma \subset \Sigma$. By the convex hull property, $\Gamma \subset \mathrm{Conv}(\gamma) \subset K$, a contradiction. □

Corollary 3.12. *Let $\Sigma^k \subset \mathbb{R}^n$ be a compact minimal submanifold and K compact convex, with $K \cap \partial\Sigma = \emptyset$. Then the inclusion $\Sigma \cap K \hookrightarrow \Sigma$ induces an injection on the $(k-1)$-st homology group.*

3.4 Monotonicity formula

Let $\Sigma^k \subset \mathbb{R}^n$ be a minimal submanifold and set $\omega_k = |B_1^k(0)|$. We denote by $B_r(x_0)$ the Euclidean ball in $\mathbb{R}^n$, and by $B_r^\Sigma(o)$ the intrinsic geodesic ball in Σ.

Theorem 3.13 (Monotonicity formula)**.** (i) *(Extrinsic version) For $x_0 \in \mathbb{R}^n$ and $0 < s < r < \mathrm{dist}(x_0, \partial\Sigma)$, assuming the intersections have finite volume, we have*

$$\frac{|\Sigma \cap B_r(x_0)|}{r^k} - \frac{|\Sigma \cap B_s(x_0)|}{s^k} = \int_{\Sigma \cap (B_r(x_0) \setminus B_s(x_0))} \frac{|(x - x_0)^\perp|^2}{|x - x_0|^{k+2}} \, \mathrm{d}\,\mathrm{vol}_\Sigma \,. \tag{1}$$

In particular,

$$\Theta_\Sigma^{\mathrm{ext}}(x_0, r) := \frac{|\Sigma \cap B_r(x_0)|}{\omega_k r^k}$$

is monotone nondecreasing. It is constant in r if and only if Σ is conical with vertex x_0 (in the relevant annulus).

If $x_0 \in \Sigma$ is a regular point, then $\Theta_\Sigma^{\mathrm{ext}}(x_0, r) \geq 1$; equality for some r holds if and only if $\Sigma \cap B_r(x_0)$ is a flat k-disk, under the usual connectedness and no-boundary assumptions.

(ii) *(Intrinsic version) For $x_0 \in \Sigma$ and $0 < s < r < \mathrm{dist}_\Sigma(x_0, \partial\Sigma)$, write $r_\Sigma(x) = \mathrm{dist}_\Sigma(x_0, x)$ and choose Euclidean coordinates so that x_0 is mapped to 0. Then*

$$\begin{aligned}\frac{|B_r^\Sigma(x_0)|}{r^k} - \frac{|B_s^\Sigma(x_0)|}{s^k} = \int_{B_r^\Sigma(x_0) \setminus B_s^\Sigma(x_0)} \frac{1}{r_\Sigma(x)^{k+1}} \Bigg[&(r_\Sigma(x) - |x|) \\ &+ \frac{|x - x_0|}{2} \left| \frac{x - x_0}{|x - x_0|} - \nu_{B^\Sigma_{r_\Sigma(x)}(x_0)}(x) \right|^2 \Bigg] \mathrm{d}\,\mathrm{vol}_\Sigma(x).\end{aligned} \tag{2}$$

Here $\nu_{B_t^\Sigma(x_0)}$ is the outward intrinsic unit normal, defined almost everywhere, and the integrand at $x = 0$ is interpreted by continuity. Thus

$$\Theta_\Sigma^{\mathrm{int}}(x_0, r) := \frac{|B_r^\Sigma(x_0)|}{\omega_k r^k}$$

is also monotone nondecreasing, with $\Theta_\Sigma^{\mathrm{int}}(x_0, r) \geq 1$. The intrinsic formula holds with the standard almost-everywhere interpretation at the cut locus [FS26].

Proof. For the extrinsic formula, we can assume $x_0 = 0$ and that Σ intersects ∂B_s and ∂B_r transversally. Consider

$$X(x) = \frac{x}{|x|^k}, \qquad DX = \frac{\mathrm{Id}}{|x|^k} - \frac{k\, x \otimes x^\flat}{|x|^{k+2}}.$$

If $\{e_i\}_{i=1}^k$ is an orthonormal basis of $T_x\Sigma$, then

$$\operatorname{div}_\Sigma X = \sum_{i=1}^k \langle D_{e_i} X, e_i\rangle = \frac{k}{|x|^k} - \frac{k|x^\top|^2}{|x|^{k+2}} = \frac{k|x^\perp|^2}{|x|^{k+2}}.$$

Integrating on $\Sigma \cap (B_r \setminus B_s)$ and applying the divergence theorem, together with $\operatorname{div}_\Sigma x = k$, gives (1). The general result follows by approximation by regular radii. If the density is constant, $x^\perp = 0$ in the annulus, hence the radial vector field is tangent to Σ and Σ is conical. Finally, if $x_0 \in \Sigma$ is regular, $\lim_{r\to 0} \Theta^{\mathrm{ext}}_\Sigma(x_0, r) = 1$.

For the intrinsic formula, assume again that $x_0 = 0$. Recall that $\Delta_\Sigma |x|^2 = 2k$, so the divergence theorem on a regular intrinsic ball yields

$$k|B^\Sigma_R| = \frac{1}{2}\int_{B^\Sigma_R} \Delta_\Sigma |x|^2 \,\mathrm{d}\,\mathrm{vol}_\Sigma = \int_{\partial B^\Sigma_R} \langle x, \nu_{B^\Sigma_R}\rangle \,\mathrm{d}\,\mathrm{vol}_{\partial B^\Sigma_R}.$$

Since

$$\langle x, \nu_{B^\Sigma_R}\rangle = R - \left[(R - |x|) + \frac{|x|}{2}\left|\frac{x}{|x|} - \nu_{B^\Sigma_R}\right|^2\right],$$

we get

$$R|\partial B^\Sigma_R| - k|B^\Sigma_R| = \int_{\partial B^\Sigma_R} \left[(R - |x|) + \frac{|x|}{2}\left|\frac{x}{|x|} - \nu_{B^\Sigma_R}\right|^2\right] \mathrm{d}\,\mathrm{vol}_{\partial B^\Sigma_R}.$$

Now use

$$\frac{\mathrm{d}}{\mathrm{d}R}\left(\frac{|B^\Sigma_R|}{R^k}\right) = \frac{R|\partial B^\Sigma_R| - k|B^\Sigma_R|}{R^{k+1}}$$

and the coarea formula to obtain (2). Since $r_\Sigma(x) \geq |x|$, its right-hand side is nonnegative. □

Remark 3.14. The monotonicity formulas also hold for immersed submanifolds, with volume counted with multiplicity. In particular, $\lim_{r\to 0} \Theta^{\mathrm{ext}}_\Sigma(x_0, r)$ is the multiplicity at x_0 (for a smooth immersion). The extrinsic formula extends to weaker notions of minimal submanifolds and is fundamental in regularity theory.

Corollary 3.15 (Florit–Simon [FS26])**.** *Let $\Sigma^k \subset \mathbb{R}^n$ be a complete, connected, smoothly immersed minimal submanifold, without boundary, and $o \in \Sigma$. Then*

$$\lim_{R\to\infty} \Theta^{\mathrm{int}}_\Sigma(o, R) = \lim_{R\to\infty} \Theta^{\mathrm{ext}}_\Sigma(o, R) \in [1, +\infty].$$

In particular, bounded intrinsic area density is equivalent to bounded extrinsic area density, and either bound implies that the immersion is proper.

Remark 3.16. The properness assumption cannot be omitted in statements about bounded complete minimal surfaces. Jorge–Xavier constructed a complete minimal immersion in $\mathbb{R}^3$ between two parallel planes [JX80], and Nadirashvili constructed a complete minimal immersion in $\mathbb{R}^3$ contained in a ball [Nad96]. Both are nonproper.

Theorem 3.17 (Extended monotonicity)**.** *Let $\Sigma^k \subset \mathbb{R}^n$ be a compact minimal submanifold with smooth boundary and $x_0 \in \mathbb{R}^n \setminus \partial\Sigma$. Define the exterior cone over $\partial\Sigma$ with vertex x_0 by*

$$E = E(x_0, \partial\Sigma) := \{x_0 + t(x - x_0) : t \geq 1,\ x \in \partial\Sigma\}, \qquad \widetilde{\Sigma} = \Sigma \cup E.$$

The volume of E is counted with multiplicity when needed. Then

$$\Theta^{\mathrm{ext}}_{\widetilde{\Sigma}}(x_0, r) := \frac{|\widetilde{\Sigma} \cap B_r(x_0)|}{\omega_k r^k}$$

is monotone nondecreasing for $r > 0$. Equality on an interval of radii forces the radial vector field to be tangent to $\widetilde{\Sigma}$ in the corresponding annulus, with vanishing boundary contribution along $\partial\Sigma$.

Proof. On the cone E, the position vector $x - x_0$ is tangent, so $\langle x - x_0, \vec{H}_E \rangle = 0$. On $\partial\Sigma$, the outward conormals of Σ and E satisfy

$$\langle x - x_0, \eta_\Sigma + \eta_E \rangle \leq 0.$$

Using the divergence theorem for $X = (x - x_0)/|x - x_0|^k$ on both pieces, the common boundary gives a nonnegative contribution to the variation of the density. The remaining term is the same nonnegative integrand as in (1). □

Theorem 3.18 (Fáry–Milnor [Far49, Mil50])**.** *Let $\Gamma \subset \mathbb{R}^3$ be a smooth simple closed curve. If*

$$\int_\Gamma |\vec{k}| \, \mathrm{d}s \leq 4\pi,$$

where $\vec{k}$ is the curvature vector, then Γ is unknotted.

Sketch. First assume $\int_\Gamma |\vec{k}| \, \mathrm{d}s < 4\pi$. Let Σ be an area-minimizing disk with boundary Γ, provided by the Douglas–Radó solution to the Plateau problem [Dou31, Rad30]. Interior regularity gives a smooth minimal immersion (and in this setting no interior branch points). It is enough to prove that Σ is embedded.

Suppose by contradiction that two sheets of Σ meet at an interior point $x_0 \in \Sigma$. Then

$$\lim_{r \to 0} \Theta^{\mathrm{ext}}_{\Sigma}(x_0, r) \geq 2.$$

Let $E = E(x_0, \Gamma)$ and $\widetilde{\Sigma} = \Sigma \cup E$. By extended monotonicity,

$$2 \leq \lim_{r \to 0} \Theta^{\mathrm{ext}}_{\widetilde{\Sigma}}(x_0, r) \leq \lim_{R \to \infty} \Theta^{\mathrm{ext}}_{\widetilde{\Sigma}}(x_0, R). \tag{3}$$

On the other hand, the compact part Σ does not contribute to the density at infinity. By the conical structure,

$$\lim_{R \to \infty} \frac{|E \cap B_R(x_0)|}{\pi R^2} = \lim_{R \to \infty} \frac{|E \cap \partial B_R(x_0)|}{2\pi R}.$$

The annulus $E \cap B_R(x_0)$ is flat away from the vertex. By Gauss–Bonnet, its two boundary components have opposite total geodesic curvature. The outer boundary contributes its length divided by R, and the contribution at Γ is bounded by the total curvature of Γ:

$$\frac{|E \cap \partial B_R(x_0)|}{R} = -\int_\Gamma \langle \vec{k}, \eta_E \rangle \, \mathrm{d}s \leq \int_\Gamma |\vec{k}| \, \mathrm{d}s.$$

Therefore

$$\lim_{R \to \infty} \Theta^{\mathrm{ext}}_{\widetilde{\Sigma}}(x_0, R) \leq \frac{1}{2\pi} \int_\Gamma |\vec{k}| \, \mathrm{d}s < 2,$$

contradicting (3). The equality case $\int_\Gamma |\vec{k}| \, \mathrm{d}s = 4\pi$ follows from the rigidity argument in the Fáry–Milnor theorem [Far49, Mil50]. □

References

[Cat42] E. Catalan, *Sur les surfaces réglées dont l'aire est un minimum*, J. Math. Pures Appl. **7** (1842), 203–211.

[Cho25] O. Chodosh, *Introduction to minimal surfaces*, 2025. URL: `https://web.stanford.edu/~ochodosh/Math286-min-surf.pdf`. Last visited on 2026/10/25. Lecture notes for MATH 286, Stanford, Winter 2025.

[CM11] T. H. Colding and W. P. Minicozzi II, *A course in minimal surfaces*, Graduate Studies in Mathematics, vol. 121, American Mathematical Society, Providence, RI, 2011.

[Dou31] J. Douglas, *Solution of the problem of Plateau*, Trans. Amer. Math. Soc. **33** (1931), no. 1, 263–321. MR1501590

[Eul44] L. Euler, *Methodus inveniendi lineas curvas maximi minimive proprietate gaudentes, sive solutio problematis isoperimetrici latissimo sensu accepti*, Marcum-Michaelem Bousquet et socios, Lausanne and Geneva, 1744.

[Far49] I. Fary, *Sur la courbure totale d'une courbe gauche faisant un nœud*, Bull. Soc. Math. France **77** (1949), 128–138. MR11,393h

[FF60] H. Federer and W. H. Fleming, *Normal and integral currents*, Ann. of Math. (2) **72** (1960), no. 3, 458–520. MR0123260

[FS26] E. Florit-Simon, *Equivalence of intrinsic and extrinsic area bounds for minimal surfaces*, 2026. Preprint, arXiv:2605.06468 [math.DG].

[Gul73] R. D. Gulliver II, *Regularity of minimizing surfaces of prescribed mean curvature*, Ann. of Math. (2) **97** (1973), no. 2, 275–305. MR0317188

[JX80] L. P. d. M. Jorge and F. Xavier, *A complete minimal surface in $\mathbb{R}^3$ between two parallel planes*, Ann. of Math. (2) **112** (1980), no. 1, 203–206. MR0584079

[Lag67] J.-L. Lagrange, *Essai d'une nouvelle méthode pour déterminer les maxima et les minima des formules intégrales indéfinies*, Œuvres de Lagrange **1** (1867), 335–362. Reprint of the original memoir published in *Miscellanea Taurinensia*, vol. 2 (1760–1761).

[Meu85] J. B. M. Meusnier, *Mémoire sur la courbure des surfaces*, Imprimerie Royale, Paris, 1785. Presented at the Académie des Sciences in 1776.

[Mil50] J. W. Milnor, *On the total curvature of knots*, Ann. of Math. (2) **52** (1950), no. 2, 248–257. MR0037509

[Mor48] C. B. Morrey Jr., *The problem of Plateau on a Riemannian manifold*, Ann. of Math. (2) **49** (1948), no. 4, 807–851. MR0027137

[MY82] W. H. Meeks III and S.-T. Yau, *The classical Plateau problem and the topology of three-dimensional manifolds: the embedding of the solution given by Douglas–Morrey and an analytic proof of Dehn's lemma*, Topology **21** (1982), no. 4, 409–442. MR0670745

[Mye41] S. B. Myers, *Riemannian manifolds with positive mean curvature*, Duke Math. J. **8** (1941), no. 2, 401–404. MR0004518

[Nad96] N. Nadirashvili, *Hadamard's and Calabi–Yau's conjectures on negatively curved and minimal surfaces*, Invent. Math. **126** (1996), no. 3, 457–465. MR1419004

[Oss70] R. Osserman, *A proof of the regularity everywhere of the classical solution to Plateau's problem*, Ann. of Math. (2) **91** (1970), no. 3, 550–569. MR0266070

[Pla73] J. Plateau, *Statique expérimentale et théorique des liquides soumis aux seules forces moléculaires*, Gauthier-Villars, Paris, 1873.

[Rad30] T. Radó, *On Plateau's problem*, Ann. of Math. (2) **31** (1930), no. 3, 457–469. MR1502955

[Sim68] J. Simons, *Minimal varieties in Riemannian manifolds*, Ann. of Math. (2) **88** (1968), 62–105.

[SY79a] R. Schoen and S.-T. Yau, *Existence of incompressible minimal surfaces and the topology of three dimensional manifolds with non-negative scalar curvature*, Ann. of Math. (2) **110** (1979), no. 1, 127–142. MR0541332

[SY79b] ———, *On the proof of the positive mass conjecture in general relativity*, Comm. Math. Phys. **65** (1979), no. 1, 45–76. MR0526976

[Web] M. Weber, *Minimal Surfaces — repository*. Online collection of minimal-surface illustrations. `https://minimalsurfaces.blog/home/repository/`. Accessed October 25, 2026.

[Whi16] B. White, *Introduction to minimal surface theory* **22** (2016), 387–438.